

Fast Search-By-Classification
for Large-Scale Databases
Using Index-Aware
Decision Trees and
Random Forests



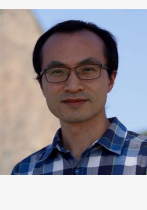
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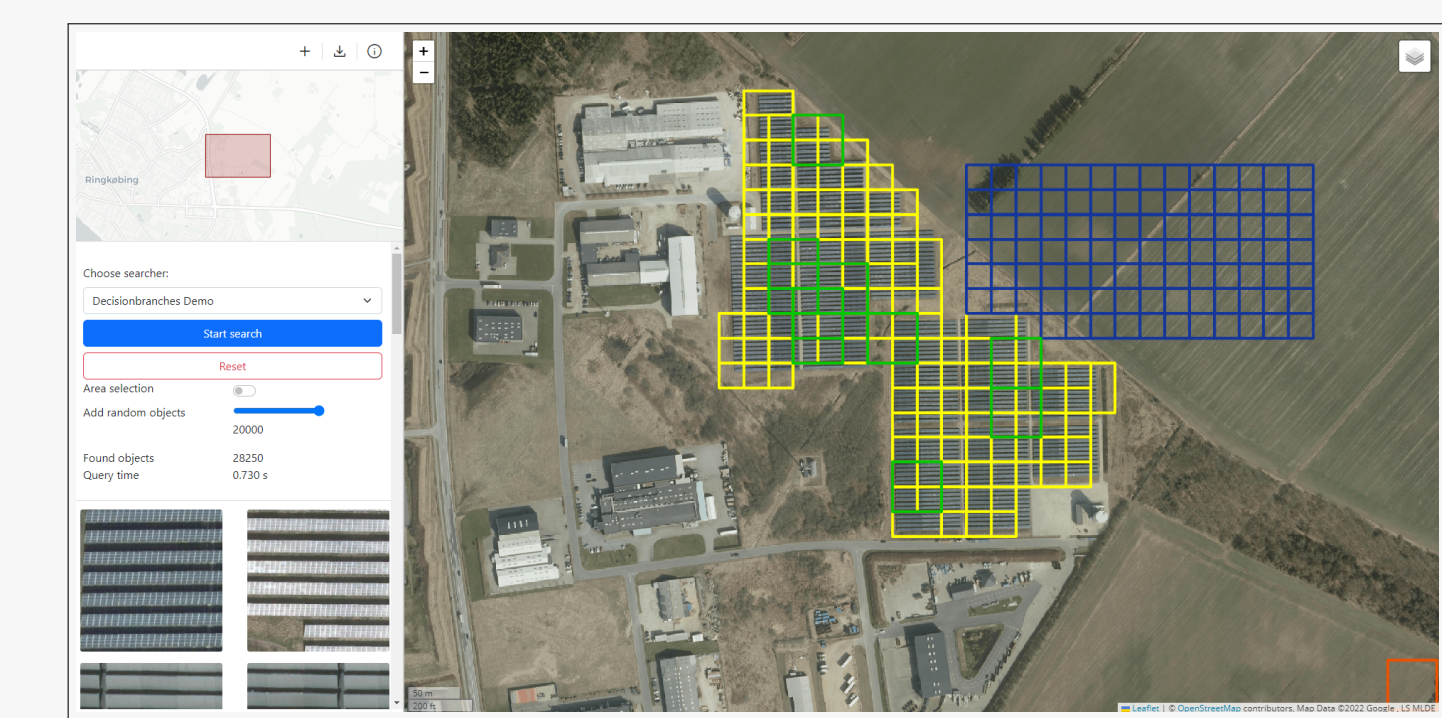
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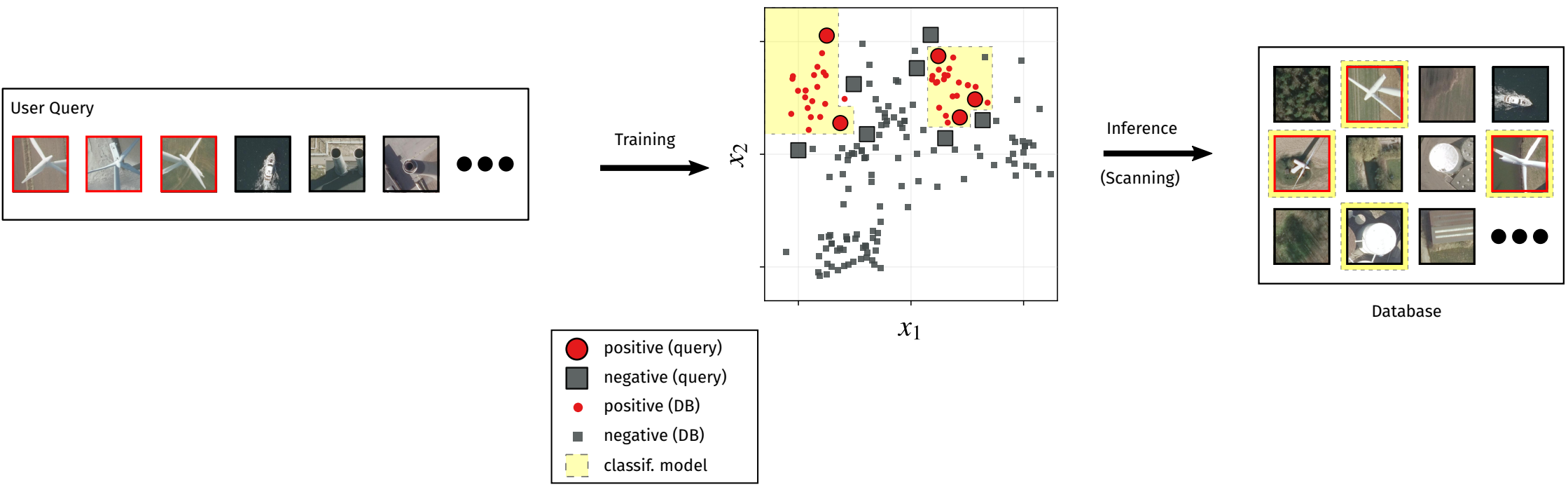
Say goodbye to tedious wait times!

Our **index-aware classifiers** make retrieving data a breeze!

Search-by-Classification: Vision and Problem

Imagine effortlessly pinpointing target items in vast databases using just a **handful of examples**. Examples of target objects are **easily** provided by users over satellite imagery, astronomy, and many other domains.

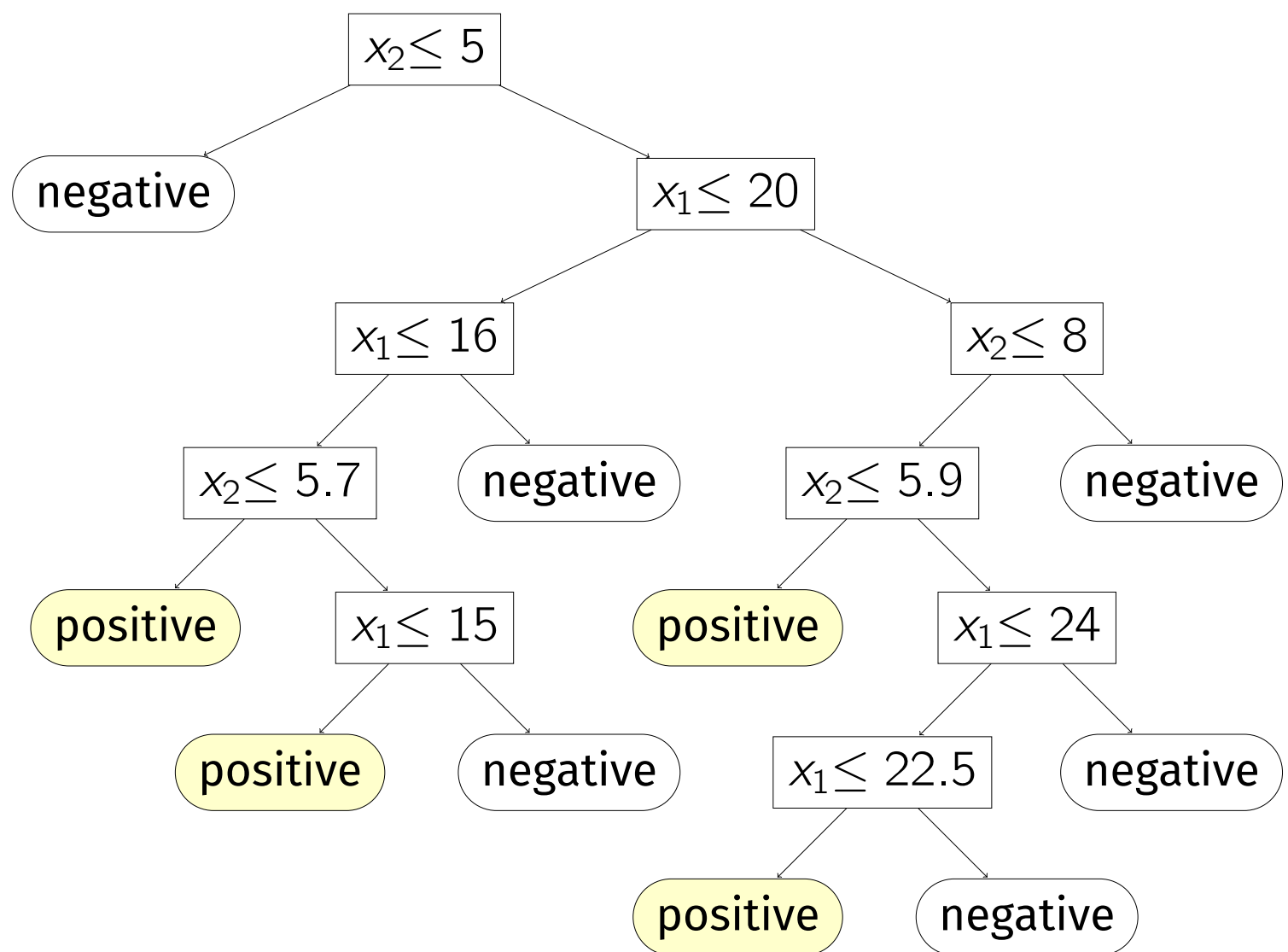
Define the user query with **positive** and **negative** examples and train an ML model with the data. **BUT** applying the learned model requires **scanning** the data. **~30 minutes** for 10^9 data points on a fast SDD.



Our idea: Use index structures to speed up inference. **No Scans needed!**

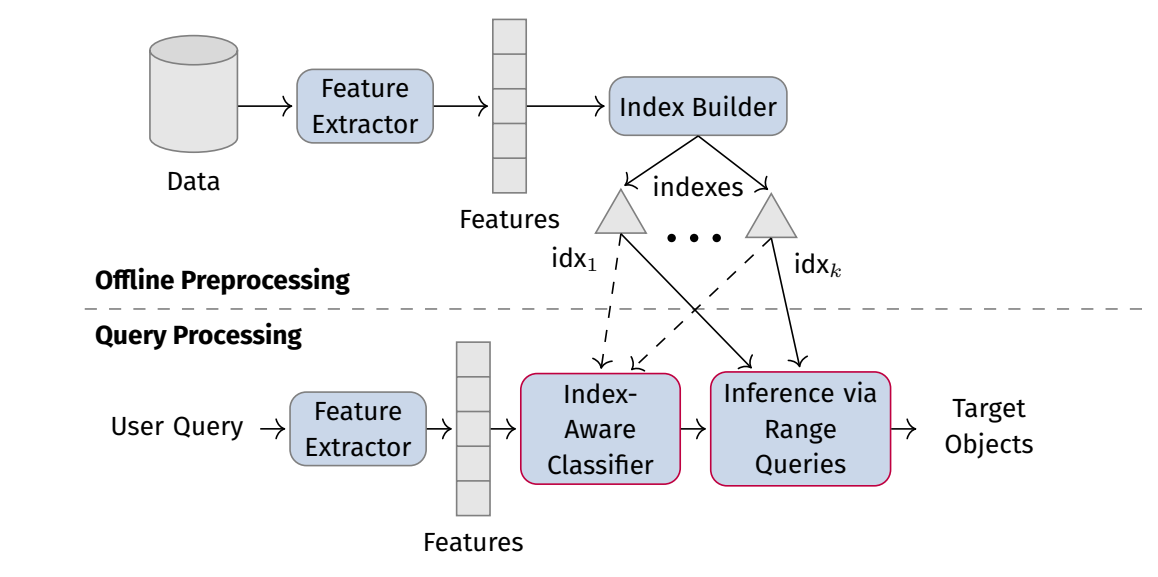
ML Model & Range Queries

Tree-based ML models can be translated into orthogonal **range queries**.

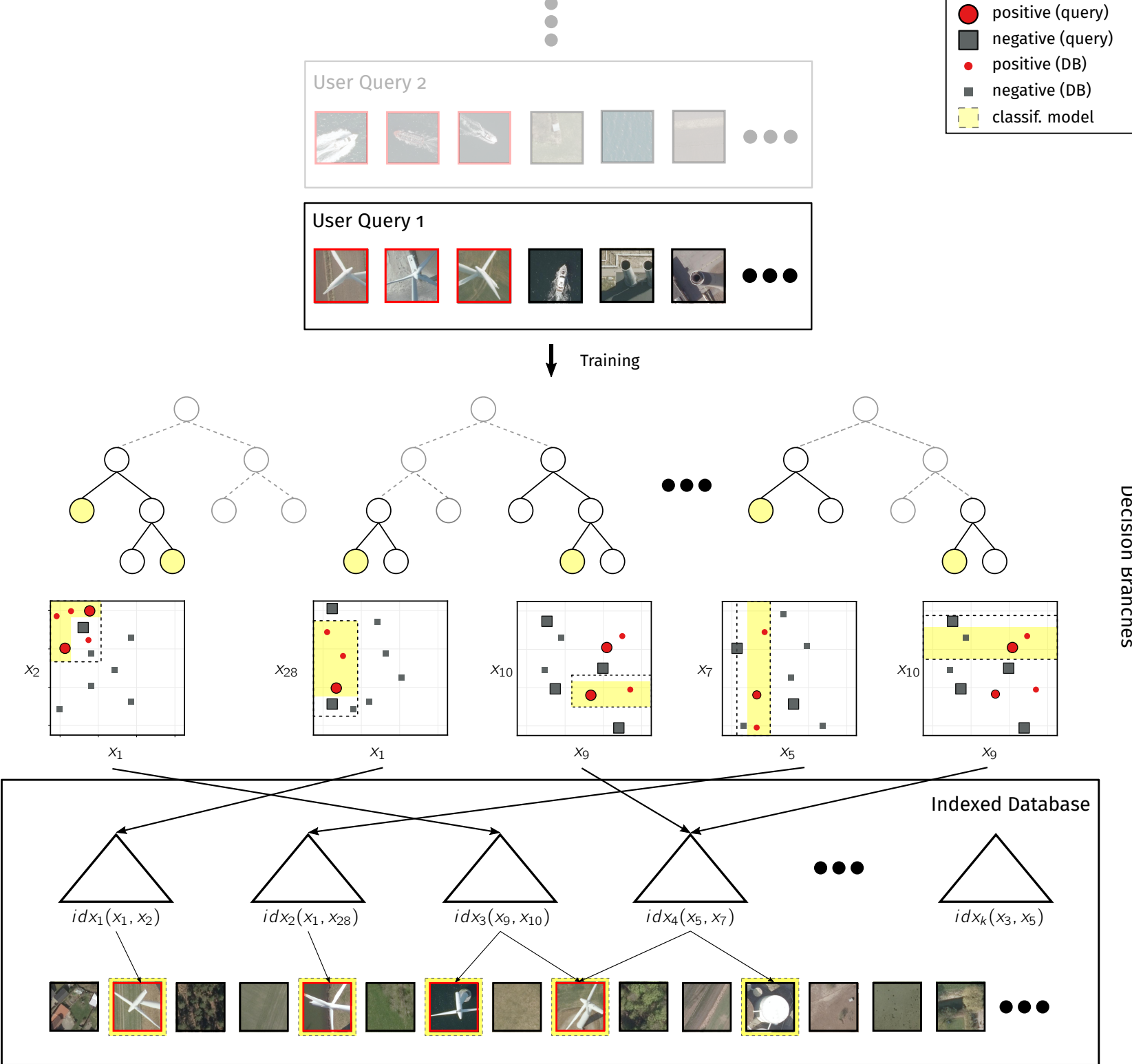


Our Approach

- Co-design** of index structures and machine learning model.
- Index-aware** construction scheme for decision trees and random forests.
- Similar classification performance** of regular tree-based models.
- Trained model can be **efficiently queried** using index structures.



Decision Branches



Index-aware Classifier

Challenge: Index structures are efficient in small feature subsets; ML models usually require many features for good performance.

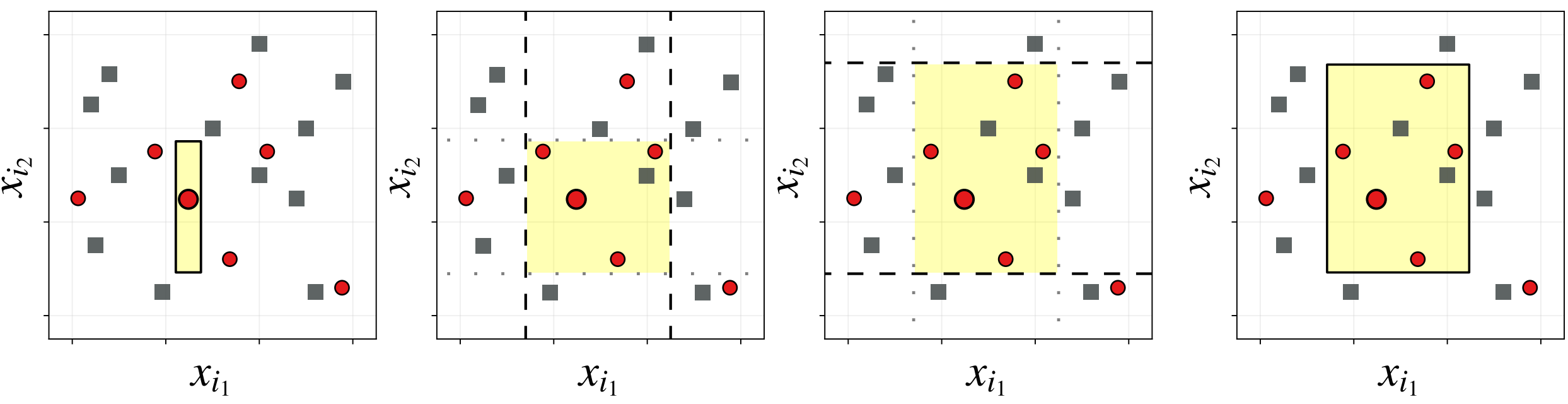
Observation: Using small, random feature subsets does not hurt classification performance.

Solution:

- Index structures are prebuilt for small, random feature subsets.
- Decision Branches are trained within these feature subspaces.
- Trained Decision Branches are translated into range queries which are supported by the pre-built index structures

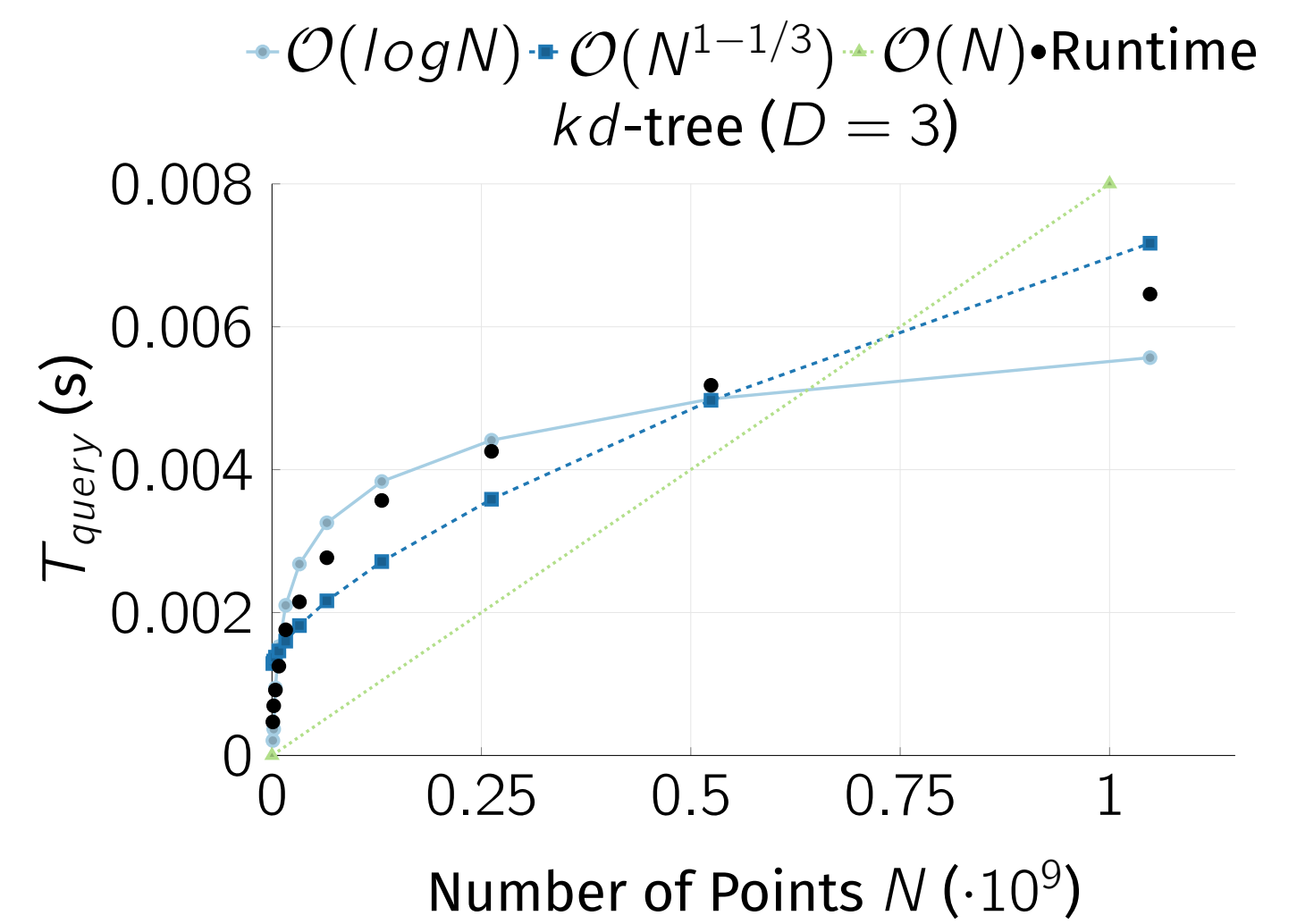
Bottom-up Construction

We propose a **bottom-up** construction scheme for decision trees that enables the efficient use of index structures. It starts from one positive point and extends its boundaries by maximizing an information gain criterion. Unlike traditional decision trees, our Decision Branches are only built in **low-dimensional feature subspaces** to be efficient with index structures while being **equally strong in classification performance**.



Scaling Behavior

Our implemented index structure falls in-between the asymptotic curves for $\mathcal{O}(\log N)$ and $\mathcal{O}(N^{1-1/3})$.



Classification Performance (F_1 -score)

DBranch (single): classification performance is **similar** to that of DTree.

DBranch ensembles: **better** classification performance than RForest.

Data set	DBranch	DTree	DBEns	RForest	ExTrees
covtype	0.355 ± .030	0.420 ± .022	0.437 ± .035	0.431 ± .027	0.449 ± .043
iris	0.972 ± .036	0.947 ± .052	0.972 ± .038	0.967 ± .027	0.977 ± .030
letter	0.585 ± .049	0.580 ± .051	0.751 ± .031	0.677 ± .054	0.735 ± .028
mnist	0.515 ± .047	0.525 ± .057	0.705 ± .026	0.612 ± .068	0.669 ± .032
satimage	0.675 ± .057	0.653 ± .053	0.775 ± .020	0.755 ± .031	0.781 ± .022
senseit	0.481 ± .034	0.512 ± .040	0.573 ± .036	0.586 ± .043	0.614 ± .033
Total	0.597 ± .042	0.606 ± .046	0.702 ± .031	0.671 ± .042	0.704 ± .031

Query Performance

DBranch (single): $\sim 700\times$ **faster**.

DBranch ensemble: $\sim 200\times$ **faster**.

Model	T_{train}	T_{query}	T_{total}	F_1 -score
DBranch	0.398	1.047	1.445	0.833
DTree	0.855	1,043.433	1,044.288	0.829
DBEns	0.993	5.666	6.658	0.914
RForest	0.274	1,319.688	1,319.961	0.904
ExTrees	0.122	1,332.026	1,332.148	0.950